



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE 24AD3107

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TITLE OF THE PROJECT	Similarity-Based Duplicate Record Detection System	
S. No.	University ID	Name
1	2420030540	Allipuram Akhil
2	2420030546	P.Ruthvik Reddy
3	2420030440	Devineni Surya
Name of the Guide		Ms.Bonagani Prathusha

Abstract:

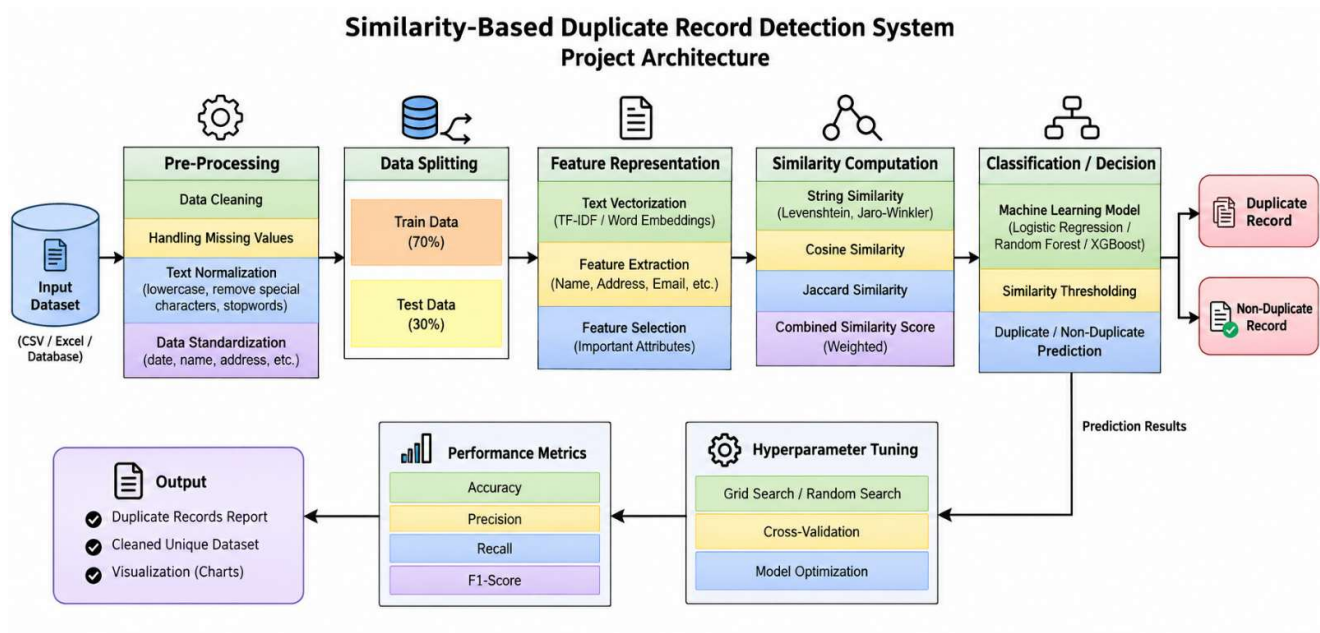
Duplicate records are a common problem in large datasets and can occur due to repeated data entry, spelling variations, inconsistent formatting, missing values, and differences in data representation. Duplicate records reduce data quality and may lead to incorrect analysis and decision-making. Identifying duplicates manually is time-consuming, especially for large datasets. Traditional methods mainly focus on exact matching and may fail to detect records with small variations. This creates a need for an automated system that can identify both exact and near-duplicate records using similarity-based techniques.

To address this need, the proposed **Similarity-Based Duplicate Record Detection System** uses data science and machine learning techniques to identify duplicate and non-duplicate records. The system accepts data from sources such as CSV, Excel, or databases and performs **data cleaning, missing value handling, text normalization, and standardization**. Important attributes such as names, addresses, email IDs, and phone numbers are extracted and converted into suitable feature representations.

The first approach introduces a String Similarity-Based Duplicate Detection technique using **Levenshtein distance and Jaro-Winkler similarity** to identify textual variations and spelling differences. **The second approach uses TF-IDF, Cosine Similarity, and Jaccard Similarity** to compare records and generate a combined similarity score. A suitable threshold is applied to classify records as duplicate or non-duplicate.

The third approach, Machine Learning-Based Duplicate Record Classification, uses algorithms such as **Logistic Regression, Random Forest, and XGBoost** to classify record pairs. **Hyperparameter tuning and cross-validation** are used to improve prediction performance.

Collectively, these approaches provide an automated and efficient solution for duplicate record detection. The system is evaluated using **accuracy, precision, recall, and F1-score** and generates a duplicate records report, cleaned dataset, and visualizations. The proposed system can be applied to **student, customer, employee, healthcare, product, and business datasets**, helping improve data quality, consistency, and reliability.



Signature of the Guide

HOD